Experiment #4: Alcohols

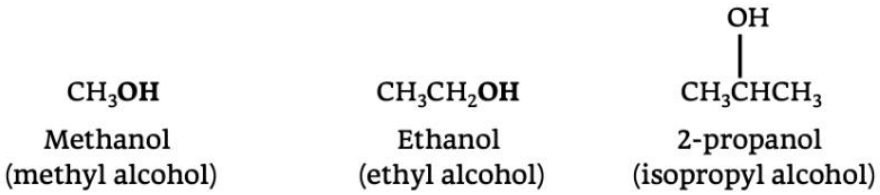
Goals

- Determine chemical and physical properties of alcohols.
- Classify an alcohol as primary, secondary, or tertiary.
- Perform a chemical test to distinguish between the classes of alcohols.
- Write the formulas of oxidation products of alcohols.

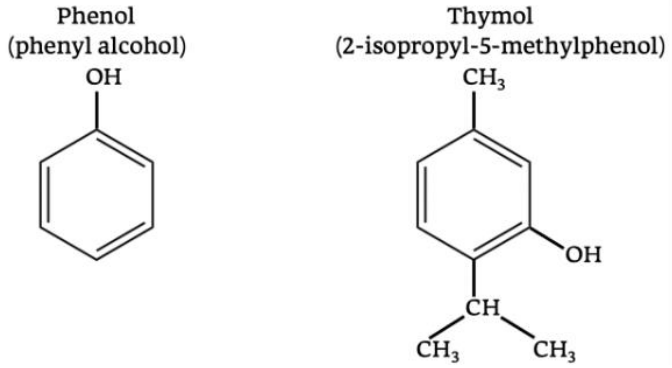
Introduction

A. Structures of Alcohols and Phenols

Alcohols are made up of the hydroxyl group (-OH). Methanol is the most basic type of alcohol. Alcoholic drinks like beer, wine, whiskey, and tequila all have ethanol in them. It is also often used in chemistry and biological labs as a preservative and a cleaner. Astringents and perfumes contain 2-propanol, which is also called rubbing alcohol or isopropyl alcohol.

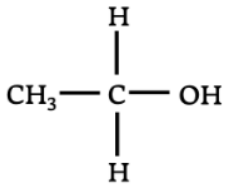


A benzene ring with a hydroxyl group is known as phenol. Concentrated solutions of phenol can be dangerous because they are caustic, can cause burns, and are carcinogenic. However, derivatives of phenol, such as thymol, are used as antiseptics and sometimes in cough drops.

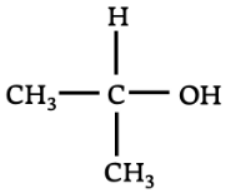


Classification of Alcohols

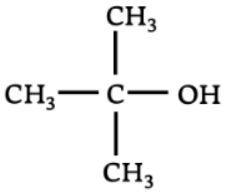
In a primary (1°) alcohol, the carbon atom attached to the -OH group is bonded to one other carbon atom. In a secondary (2°) alcohol, it is attached to two carbon atoms, and in a tertiary (3°) alcohol to three carbons. See Examples listed below.



Ethanol
primary (1°) alcohol



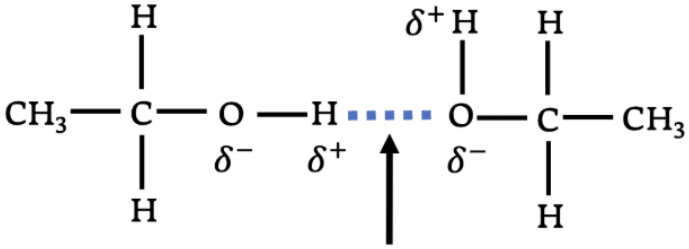
2-propanol
secondary (2°) alcohol



2-methyl-2-propanol
tertiary (3°) alcohol

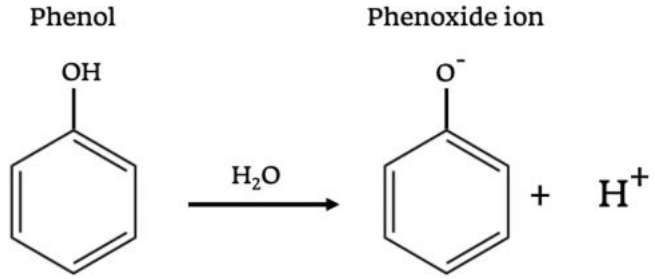
B. Properties of Alcohols and Phenols

The polarity of the hydroxyl group (-OH) makes alcohols with four or fewer carbon atoms soluble in water because they can form hydrogen bonds. However, in longer carbon alcohols, a large hydrocarbon section makes them insoluble in water



Hydrogen Bonding
(Based on polarity and electronegativity of functional groups)

Acidity and Aromaticity of Phenol

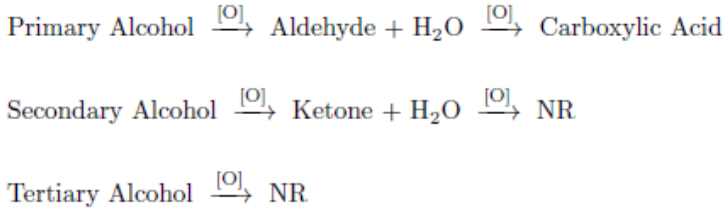


In water, phenol acts as a weak acid because the hydroxyl group ionizes slightly. Although phenol has six carbon atoms, its acid behavior, which creates an ion, makes it more soluble in water. Aromatic compounds like phenol are rigid and planar structures that pack closely together, increasing the intermolecular forces between them, and therefore increasing properties such as boiling points. The aromatic nature in which electrons are delocalized around the ring through resonance contributes to greater stability, which makes them less reactive.

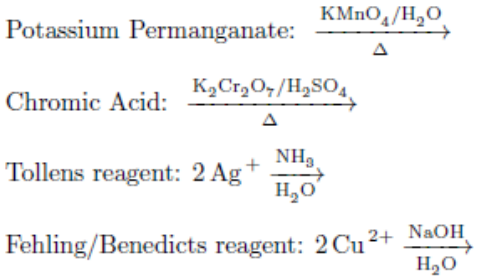
C. Oxidation of Alcohols

Alcohols can undergo two main chemical reactions: dehydration and oxidation. Classes of alcohols can be identified by utilizing different oxidizing reagents, such as potassium permanganate, chromic acid, Tollens reagent, and Benedicts reagent. Primary and secondary alcohols will oxidize, but tertiary alcohols do not react due to the lack of available hydrogen.

General Oxidation Reactions of Alcohols:



Oxidation Tests for Alcohols:



Today, you will use the permanganate test, which is a general, non-specific test that gives a positive result for multiple functional groups. You have previously used this test to distinguish between alkanes and alkenes. A primary or secondary alcohol will oxidize to give a positive result, producing a brown color change and/or brown precipitate. Tertiary alcohols cannot be oxidized and will give a negative result, remaining purple in color.

The chromic acid test can be used when a more specific test is needed. Another useful test is the Lucas test, which is a substitution reaction using zinc chloride. The rate of reaction gives a clear distinction between all classifications of alcohols, with tertiary alcohols as the most reactive.

Hazards & Safety

Related Hazards & Safety Instructions

- **Strong Oxidizer**: Potassium permanganate is a strong oxidizer and is violently reactive with alcohols in the presence of concentrated acids. Potassium permanganate ($KMnO_4$) will cause staining upon contact with skin.

- **Flammability:** The organic compounds used in this lab are flammable and present a significant fire hazard. Do not use burners or any open flames in the lab while working with these flammable organic compounds.

- **Burns:** The oxidation of alcohols can be an exothermic process and may give off heat. Hot plates and heated glassware need to be adequately cooled before handling.

- **Odor Detection:** Do not inhale directly from the tubes. To check an odor, hold your breath and gently fan (waft) the fumes from the top of the test tube toward you.

Waste Instructions

- All chemical waste should be disposed of in non-halogenated waste.

Procedure

Instructor Note: Start a hot water bath in a hood immediately so that it is ready for Part C. Monitor heat to maintain at 60°C.

A. Types of Hydrocarbons with Model Kits

Using the organic model kits, observe and construct models of ethanol, 2-propanol, and *t*-butyl alcohol (2-methyl-2-propanol). Classify each alcohol as a primary, secondary, or tertiary alcohol. Write the skeletal formula of each as well as phenol.

B. Properties of Alcohols

Materials: 5 test tubes, pH paper, stirring rod, ethanol, 2-propanol, t-butyl alcohol (2-methyl-2-propanol), cyclohexanol, and your choice of unknown

1. **Odor:** Place 5 drops of each of the alcohols and your chosen unknown to five separate test tubes. Carefully detect the odor of each by safely wafting. Record your observations.
2. **Acidity:** Obtain a container of pH paper. For each sample, use a glass stir rod to dip into the test tube and touch a drop to all squares on the pH paper. Compare the resulting color combination of the pH paper with the chart on the container to determine the pH of each sample. Record the pH data in your Report Sheet.

3. **Solubility in Water:** Add 2 mL of water (measure with plastic pipet or about 2 finger-widths) to each test tube. Shake and determine whether each alcohol is soluble. If the substance is soluble in water, you will see a clear solution with no separate layers. If it is insoluble, a cloudy mixture or separate layers will form. Record your observations.

C. Oxidation of Alcohols

Materials: 5 test tubes, ethanol, 2-propanol, t-butyl alcohol (2-methyl-2-propanol), cyclohexanol, unknown, 0.1% potassium permanganate solution.

1. Place 8 drops of each alcohol and your chosen unknown into separate test tubes. Carefully add 2 drops of permanganate solution to each. Watch for a color change or the appearance of brown precipitate. In your Report Sheet, note any visible changes and the approximate rate of reaction (time) for positive test results.

2. After 10 minutes, place any unreacted test tubes into a warm water bath and heat for 3 minutes. Carefully remove the tubes and record your observations, noting that heat was necessary by including the delta sign, Δ, for these samples.

3. Draw the skeletal formula of each alcohol.

4. Classify each alcohol as primary (1°), secondary (2°), or tertiary (3°).

5. Draw the skeletal formula of the products where oxidation occurred. When there is no change in color, no oxidation took place. Write "No Reaction" or NR.

Identification of Unknown Substance

Use the test results to identify the unknown that you choose as one of the four compounds used throughout the experiment.

Report Sheet

Name _____ Date _____

Lab Partner _____ Section _____

A. Structures of Alcohols and Phenols

ethanol	**2-propanol**
Classification:	Classification:
***t*-butyl alcohol (2-methyl-2-propanol)**	**phenol**
Classification:	

B. Properties of Alcohols

Alcohol	B.1 Odor	B.2 pH	B.3 Solubility
Ethanol			
2-propanol			
t-butyl alcohol			
Cyclohexanol			
Unknown ___			

C. Oxidation of Alcohols

Alcohol	C.1 $KMnO_4$ Result	C.2 Structure	C.3 Classification	C.4 Oxidation Product
Ethanol				
2-propanol				
t-butyl alcohol				
Cyclohexanol				
Unknown ___				

Identification of Unknown Substance **Unknown Letter _____**

Summary of Testing	Results	Conclusions/Match
B.1 Odor		
B.2 pH		
B.3 Solubility		
C. Oxidation $KMnO_4$		
Name of Unknown	**Structure**	

Questions and Problems

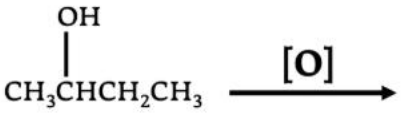
Q.1 Write the skeletal structures and classifications of the following alcohols:

1-pentanol	3-pentanol
Cycloheptanol	1-methylcyclopentanol

Q.2 Look up and draw the skeletal structures and list uses of thymol, menthol, and resorcinol. Circle the phenol functional group in each.

Q.3 Write the products of the following reactions (if no reaction, write NR):

A. $CH_3CH_2CH_2OH \xrightarrow{[O]}$

B. 

C. 